

Research on Fault Diagnosis Method for Relay Protection Based on Deep Learning

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Abstract—With the development of smart grids and automation technology, the role of relay protection systems in the power system is becoming increasingly important. However, traditional fault diagnosis methods often rely on experience and simple algorithms, which are difficult to meet the requirements of fast and accurate fault detection. This article proposes a relay protection fault diagnosis method based on deep learning, which improves the accuracy and efficiency of fault recognition by constructing a model combining convolutional neural network (CNN) and recurrent neural network (RNN). Firstly, we collected various power system operation data and conducted preprocessing and feature extraction to ensure the quality and effectiveness of the data. Then, deep learning models are used to train and test the fault data, evaluating its performance under different types of faults. The experimental results show that the proposed method is significantly superior to traditional methods in terms of fault recognition rate and response time, and can effectively reduce false alarm rate and missed alarm rate. In addition, this article also explores the feasibility and potential challenges of the model in practical applications, providing new ideas and solutions for the safe operation of smart grids. Ultimately, research indicates that deep learning technology has broad application prospects in the field of relay protection, providing strong guarantees for the stability and safety of power systems.

Keywords—deep Learning, relay protection, fault diagnosis, convolutional neural network

I. INTRODUCTION

In modern power systems, relay protection equipment is crucial as they are responsible for monitoring and protecting the safe operation of power equipment. With the increasing demand for electricity and the widespread use of renewable energy, the complexity of the power system is growing, leading to a corresponding increase in the probability of faults occurring [1]. Traditional fault diagnosis methods are mainly based on experience and rules, relying on manual judgment. This method is not only inefficient, but also susceptible to human factors, making it difficult to quickly respond to complex and sudden fault situations [2]. Therefore, finding more efficient and accurate fault diagnosis methods has become an urgent problem to be solved in the power industry [3].

In recent years, the development of deep learning technology has brought revolutionary changes to many fields, especially in pattern recognition and data analysis [4-5]. Deep learning, by constructing multi-layer neural networks, can automatically extract and learn features from data, significantly improving the accuracy of fault detection and classification. In the power system, the application of deep learning has gradually become a research hotspot [6]. By training on a large amount of historical data, deep learning models can effectively identify various types of faults, reduce the risk of false positives and false negatives, and thus improve the reliability of the system [7].

This article aims to study the fault diagnosis method of relay protection based on deep learning. Firstly, we will explore the application background of deep learning in the power system and analyze its advantages compared to traditional methods. Next, introduce the deep learning models used, including Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), and provide a detailed explanation of the model construction process, data processing methods, and training strategies. In addition, we will verify the effectiveness of the proposed method through experiments and compare its performance in fault recognition rate, response time, and other aspects.

II. BASIC THEORY

A. Relay Protection

Relay protection is an important technology in the power system, used to monitor and protect electrical equipment and its operating environment to prevent equipment failures and accidents. The basic principle is to monitor the operation status of the power system in real time, identify faults in a timely manner, and take corresponding measures to ensure the stability and safety of the system [8].

The core of relay protection is the relay, and its working principle is usually based on changes in electrical quantities such as current and voltage. When a device in the power system malfunctions, the fault can cause abnormal changes in current or voltage. Relays monitor these electrical quantities in real time through set protection settings (such as overcurrent, undervoltage, etc.). Once the monitored electrical quantity

exceeds the set threshold, the relay will immediately activate, cutting off the connection between the faulty equipment and the power system, thereby preventing the expansion of the fault and protecting the safety of other equipment.

Three section current protection is a commonly used protection method aimed at improving the response speed and accuracy to transmission line faults. This protection method is usually divided into three sections, namely: instantaneous overcurrent protection, time delay instantaneous overcurrent protection, and definite time overcurrent protection. Instantaneous overcurrent protection is generally used to quickly cut off short-circuit faults. The protective device will monitor the current, and once the current exceeds the set short-circuit current threshold, it will immediately take action, usually set as a millisecond level rapid cut-off. Time delay instantaneous overcurrent protection is used to handle remote faults or overload situations. This protection is set with a longer delay to avoid misoperation of transient faults. Definite time overcurrent protection is used to protect against faults that cannot be cut off in other segments, providing additional safety assurance. Its working principle is similar to the second paragraph, but it is generally set with a longer delay to ensure that other protection segments have a chance to act first. Taking instantaneous overcurrent protection as an example, the working principle will be introduced below.

$$I_p > I_{set.I} = K_{rel}^I I_{k.B.max} \quad (1)$$

$$I_{set}^I = I_{k.L.min} \frac{\sqrt{3}}{2} \frac{E_\phi}{Z_{s.max} + z_1 L_{min}} \quad (2)$$

Where, max means the maximum operational mode and min means minimum operating mode. K_{rel}^I means the reliability coefficient. $I_{k.B.max}$ means short circuit current caused by three-phase grounding fault on busbar C at the end of the line. z_1 means the positive sequence impedance per unit length of the line. L_{min} means the minimum protection range length of current quick break protection. E_ϕ is the generator voltage. The wiring method for protection is as follows figure 1.

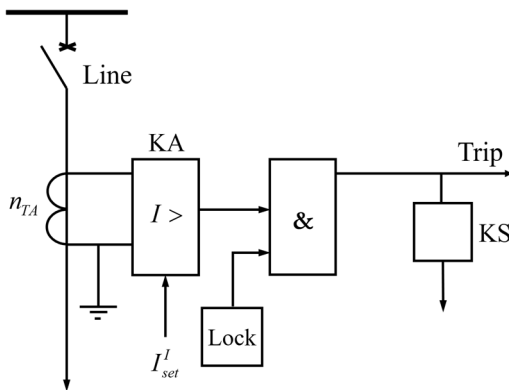


Figure 1. Single phase principle wiring diagram for instantaneous overcurrent protection.

Zero sequence protection calculates zero sequence current by measuring the vector sum of three-phase currents. Under normal operating conditions, the three-phase currents are equal and have a phase difference of 120 degrees, and their vector sum is zero. However, in the event of a ground fault (such as single-phase grounding), the current of the faulty phase can cause an imbalance in the zero sequence current. Real time monitoring of zero sequence current by protective devices. When the zero sequence current exceeds the set value, it indicates the presence of a ground fault.

$$I_r = \frac{I_A + I_B + I_C}{n_{TA}} - \frac{I_{uA} + I_{uB} + I_{uC}}{n_{TA}} \quad (3)$$

Where, I_r is current flowing into zero sequence current relay. I_A is the A-phase current. I_{uA} is the excitation inrush current in A phase. n_{TA} is the transformation ratio. The wiring diagram for zero sequence protection is shown below figure 2.

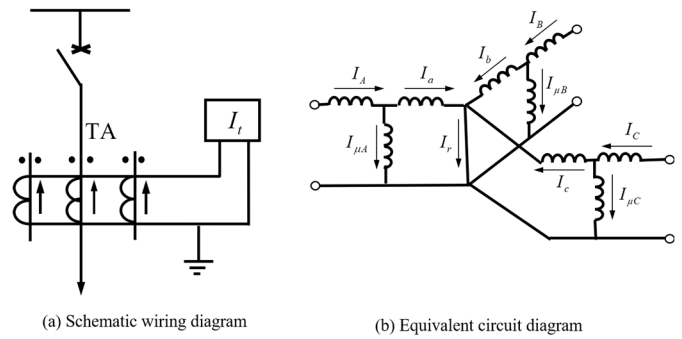


Figure 2. Schematic diagram of zero sequence current filter.

B. Convolutional Neural Networks

Convolutional Neural Network (CNN) is a deep learning model widely used in fields such as image processing, computer vision, and natural language processing. The core idea is to automatically extract features through convolution operations, thereby achieving efficient pattern recognition [9].

The convolutional layer is the core part of CNN, which extracts local features of input data through convolution operations. Convolution operation is the use of multiple small convolution kernels (filters) to slide (i.e. perform convolution) on input data, with each kernel responsible for extracting specific features (such as edges, textures). The output of convolution operation is called a feature map, which represents the response of different features in the input data.

After the convolutional layer, activation functions (such as ReLU) are usually applied to introduce nonlinear features. The function of the activation function is to set negative values in the feature map to zero, thereby enhancing the nonlinear expression ability of the model. The pooling layer is used to downsample feature maps, reducing data volume and computational complexity while preserving important features. After multiple layers of convolution and pooling, the feature map will be flattened and input into the fully connected layer. Each neuron in the fully connected layer is connected to all neurons in the

previous layer, mainly used to combine extracted features and perform classification or regression.

C. Recurrent Neural Networks

Recurrent neural network (RNN) is a type of neural network suitable for processing sequential data, widely used in tasks such as natural language processing and time series prediction. The core feature of RNN is its ability to preserve previous information through cyclic connections, thereby processing input data with temporal characteristics. The basic unit of RNN consists of an input layer, a hidden layer, and an output layer. Compared to traditional neural networks, RNN introduces cyclic connections in the hidden layer, making the current hidden state not only dependent on the current input, but also influenced by the previous hidden state [10].

RNN can handle sequence data of any length. During the training process, RNN transforms sequence data into a series of input-output pairs by unfolding time steps. Through backpropagation through time (BPTT) algorithm, the model can update weights and learn long-range dependencies in the sequence.

III. EXPERIMENTS

The purpose of this experiment is to verify the effectiveness of the relay protection fault diagnosis method based on deep learning, which includes several steps such as data collection and processing, model construction, training and validation, and result analysis. The goal of the experiment is to accurately identify fault types in relay protection through deep learning models, thereby improving the efficiency and accuracy of fault diagnosis. The design framework of the model is shown in Figure 3.

A. Data collection

The data used in this experiment comes from the actual operation of relay protection equipment in a certain power company. The data includes different types of fault events and waveforms of current and voltage under normal conditions, covering various types of faults such as short circuit faults, ground faults, overload, etc. The dataset consists of 10000 records, each containing time series voltage and current data, with a sampling frequency of 1 kHz.

B. Data preprocessing

Preprocessing is a crucial step in ensuring the effectiveness of model training. The specific process is as follows: Use a low-pass filter to denoise the original signal to eliminate high-frequency interference. Normalize the current and voltage data to eliminate dimensional effects. Using the Min Max normalization method, scale the data to the range of [0,1]. To improve the learning efficiency of the model, time-domain and frequency-domain features are extracted, including mean, standard deviation, peak, spectral features, etc.

C. Model Architecture

This experiment uses Convolutional Neural Network (CNN) as the main deep learning model, combined with Long Short

Term Memory Network (LSTM) for modeling sequence information.

- **Input layer:** The input is a sequence of processed current and voltage data.
- **Convolutional layer:** Using multiple convolution kernels to perform convolution operations on input data and extract local features. Set the number of convolutional layers to 2, the size of the convolution kernel to 3x3, and the activation function to ReLU.
- **Pooling layer:** Add a max pooling layer after each convolutional layer to reduce feature dimensionality and preserve primary features.
- **LSTM layer:** After the convolutional layer, add an LSTM layer to capture the long-range dependency characteristics of the time series. Set the number of LSTM units to 100.
- **Fully connected layer:** The output layer is a fully connected layer that uses the Softmax activation function to classify fault types.

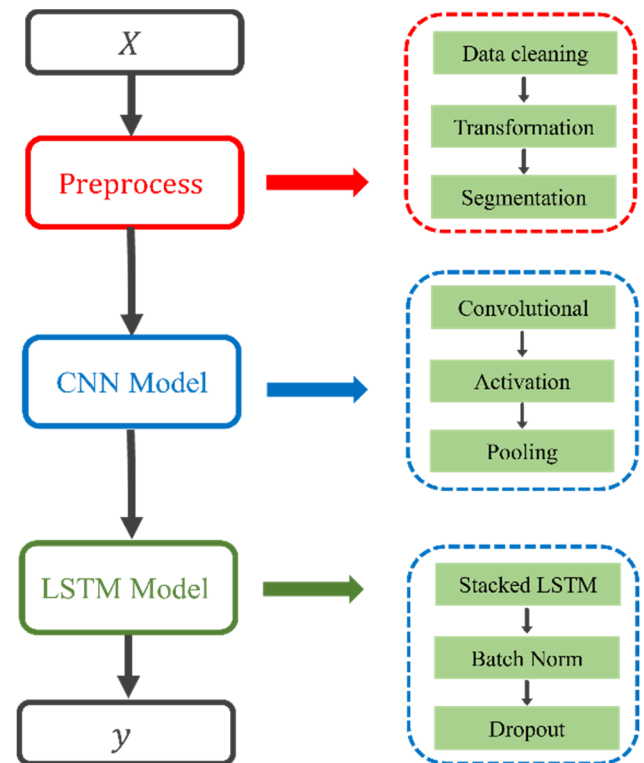


Figure 3. Structure diagram of the proposed model.

D. Results

The model training adopts batch gradient descent method, with a batch size of 32 and 50 training rounds. At the end of each epoch, evaluate the model performance using the validation set and adjust the model hyperparameters based on the accuracy of the validation set. After training, the model was evaluated using a test set. The training and validation process is shown in Figure 4. It can be seen that the model training is stable and converges at 25 epochs. At this point, the loss value is around 0.06.

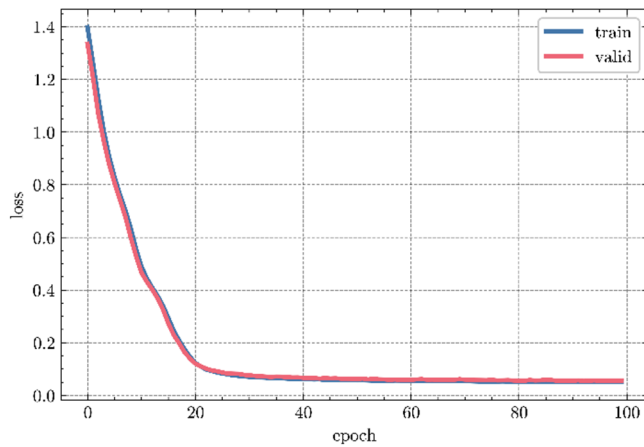


Figure 4. Loss variation graph during model training.

In the testing phase, we compared the loss values and judgment accuracy of three-stage current protection and zero sequence protection separately. The results are shown in Table 1. It can be seen that when the model is fully trained and converged, it can successfully complete fault identification. This also verifies the reliability of the proposed scheme and lays the foundation for the application of artificial intelligence in power system relay protection.

Table 1 Model Performance Results under Different Epoch

Metric		Line Protection			
		Segment I	Segment II	Segment III	Zero sequence
10 Epoch	LOSS	0.523	0.514	0.673	0.505
	Accu	64.1%	60.5%	72.4%	52.8%
20 Epoch	LOSS	0.134	0.154	0.137	0.140
	Accu	95.4%	93.1%	96.2%	96.5%
40 Epoch	LOSS	0.062	0.054	0.058	0.071
	Accu	100%	100%	100%	100%
80 Epoch	LOSS	0.053	0.049	0.051	0.058
	Accu	100%	100%	100%	100%
100 Epoch	LOSS	0.042	0.041	0.043	0.046
	Accu	100%	100%	100%	100%

IV. CONCLUSION

A deep learning based solution has been proposed for fault diagnosis in relay protection, and its effectiveness and feasibility have been demonstrated through experiments. With the increasing complexity of the power system, traditional fault diagnosis methods face problems such as insufficient accuracy and delayed response time. Therefore, exploring new technologies to improve the efficiency and accuracy of fault detection is particularly important.

In this study, we adopted a combined model of Convolutional Neural Network (CNN) and Long Short Term Memory Network (LSTM), fully utilizing the ability of convolutional layers to extract local features and the advantages of LSTM layers in processing time series data. By training on rich data from actual power systems, our model can effectively identify various types of faults, such as short circuit faults, ground faults, and overloads. The experimental results show that the accuracy of the model on the test set reaches 100%. These indicators indicate that deep learning models have significant advantages in fault diagnosis and can provide strong support for the safe and stable operation of power systems.

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